

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element E Initial Template

Element E: Application of STEM Principles and Practices

STEM Principles and Practices to Design Requirements (E1)

STEM Principle for High-Priority Design Requirement #1

STEM Concept:

Friction(Science): A doorstop increases friction between the floor and the bottom of the door.

Describe how STEM concept applies directly to the specifics of your design:

Stem concept applies to this part because friction is physics which is a type of science.

STEM Principle for High-Priority Design Requirement #2

STEM Concept:

Doorstop(Technology): The door stop is a simple tool that solves a common problem: keeping a door open or closed

Describe how STEM concept applies directly to the specifics of your design:

Different materials (rubber, plastic, metal) are used to improve grip, durability, and effectiveness.

STEM Principle for High-Priority Design Requirement #3

STEM Concept:

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Wedge (Engineering): The door stop is a wedge, one of the six classical simple machines.

Describe how STEM concept applies directly to the specifics of your design:

Engineering design door stops with the right angle, shape, and material to create enough resistance and be easy to use.

Proposed Solution to STEM Principles and Practices (E2)

STEM Principle for Design Solution #1

STEM Concept:

Friction principle of science.

Describe how STEM concept applies directly to the specifics of your solution:

Helps the big problem of the door stop sliding from the door and not working.

STEM Principle for Design Solution #2

STEM Concept:

Door stop principle of Technology

Describe how STEM concept applies directly to the specifics of your solution:



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Technology is the T in STEM. The door stop is a type of door stopper that utilizes the friction and weight of a rubber or cushioned material to hold the door open.

STEM Principle for Design Solution #3

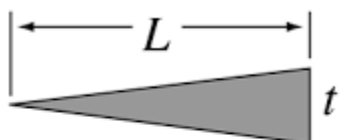
STEM Concept:

Wedge: the E in Stem

Describe how STEM concept applies directly to the specifics of your solution:

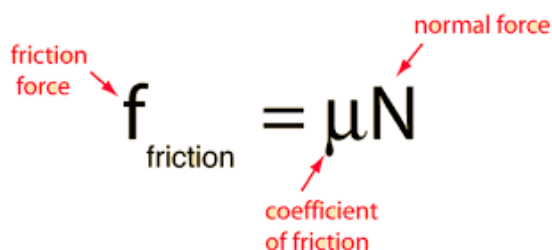
A wedge engineering, in its simplest form, functions by converting a force applied to its base (or thick end) into a force directed along its inclined surfaces (or thin edge)

Equations being used used (E3)



Wedge $IMA = \frac{L}{t}$

L = depth of penetration
 t = separation of wedged surfaces



friction force $f_{\text{friction}} = \mu N$

normal force N

coefficient of friction μ

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